

AI ASSISTANT CODING ASSIGNMENT-14.2

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Batch: 33

Semester:VI

Branch:CSE(AIML)

Lab14:WebDesignApplication–AI-AssistedHTML/CSS/JSGeneration Lab

Objectives

- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

TaskDescription1–(LandingPageDesign)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

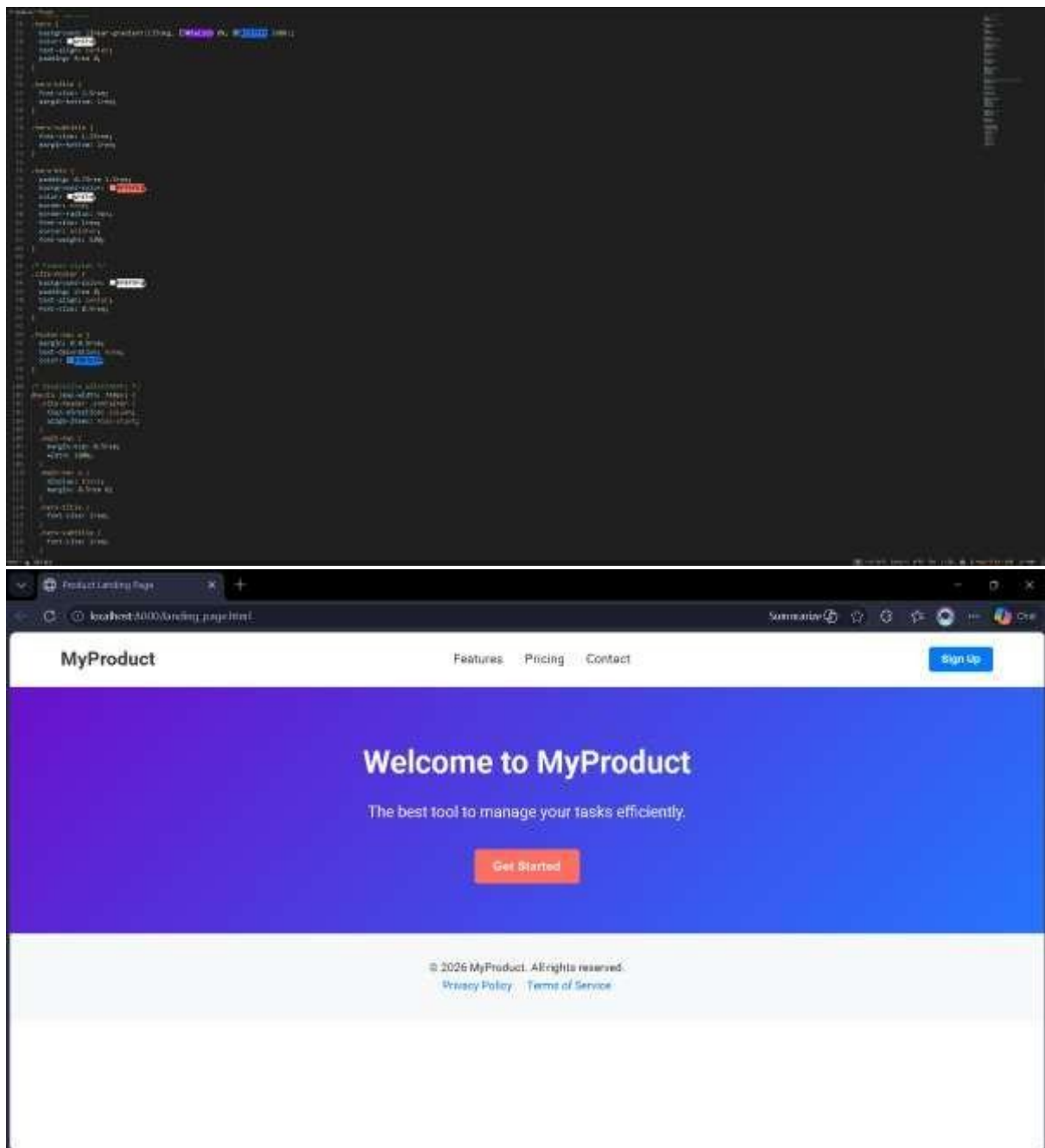
```

1 # Import the module
2 from tkinter import *
3
4 # Create a window
5 root = Tk()
6 root.title("Tkinter Window")
7 root.geometry("400x400")
8
9 # Create a label
10 label = Label(root, text="Hello, Tkinter!")
11 label.pack()
12
13 # Create a button
14 button = Button(root, text="Click Me")
15 button.pack()
16
17 # Define a function
18 def click():
19     label.config(text="You clicked!")
20
21 # Bind the button click event to the function
22 button.config(command=click)
23
24 # Run the main loop
25 root.mainloop()

```

```

// main class
// 1. first round: all types are 0
// 2. weight: 0
// 3. padding: 0
// 4. non-linear: false, true
//
// 5. bias
// 6. first round: "initial", last round: "final"
// 7. loss: "l1", "l2", "l3", "l4", "l5"
// 8. color: 0x000000
//
// 9. iteration
// 10. alpha: 0.001
// 11. max-iter: 100000
// 12. margin: 0.001
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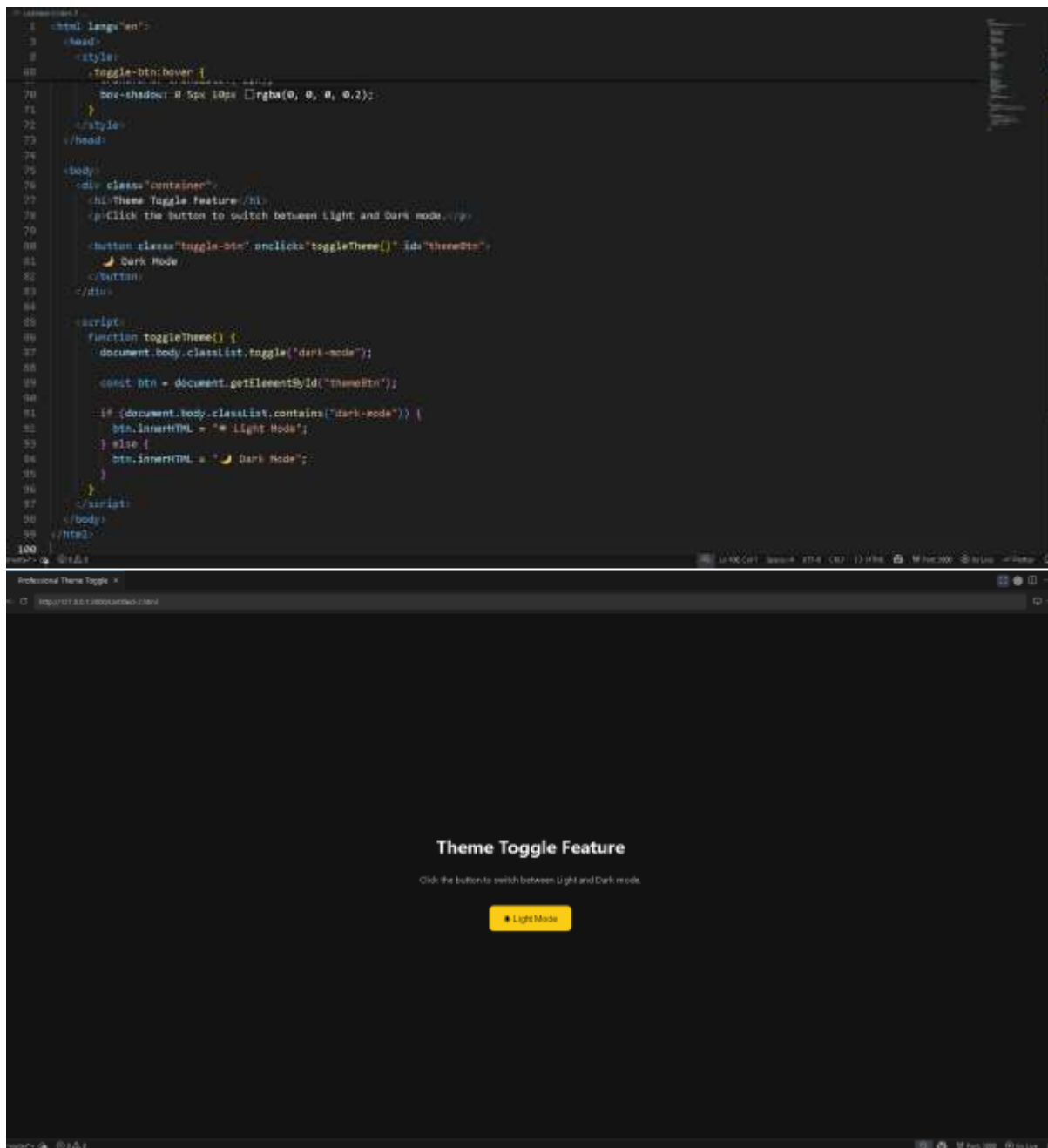
TaskDescription2–(ThemeToggleFeature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use A11y to manage color contrast and accessibility.
- Ensure the theme preference applies instantly on click.

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Professional Theme Toggle</title>
7
8 <style>
9 /* CSS Variables for Themes */
10 :root {
11   --bg-color: #ffffff;
12   --text-color: #002222;
13   --button-bg: #2e8b57;
14   --button-text: #ffffff;
15 }
16
17 body.dark-mode {
18   --bg-color: #121212;
19   --text-color: #f5f5f5;
20   --button-bg: #ffeb3b;
21   --button-text: #000000;
22 }
23
24 /* Page Styling */
25 body {
26   margin: 0;
27   font-family:
28     Segoe UI,
29     Arial,
30     sans-serif;
31   background: var(--bg-color);
32   color: var(--text-color);
33   transition:
34     background 0.4s,
35     color 0.4s;
36 }
37
38 /* Container */
39 .container {
40   display: flex;
41   flex-direction: column;
42   justify-content: center;
43   align-items: center;
44   height: 100vh;
45   text-align: center;
46 }
47
48 /* Heading */
49 h1 {
50   margin-bottom: 10px;
51 }
52
53 /* Toggle Button */
54 .toggle-btn {
55   margin-top: 20px;
56   padding: 10px 25px;
57   font-size: 16px;
58   border: none;
59   border-radius: 5px;
60   background: var(--button-bg);
61   color: var(--button-text);
62   cursor: pointer;
63   transition:
64     transform 0.2s,
65     box-shadow 0.2s;
66 }
67
68 .toggle-btn:hover {
69   transform: translateY(-2px);
70 }
```



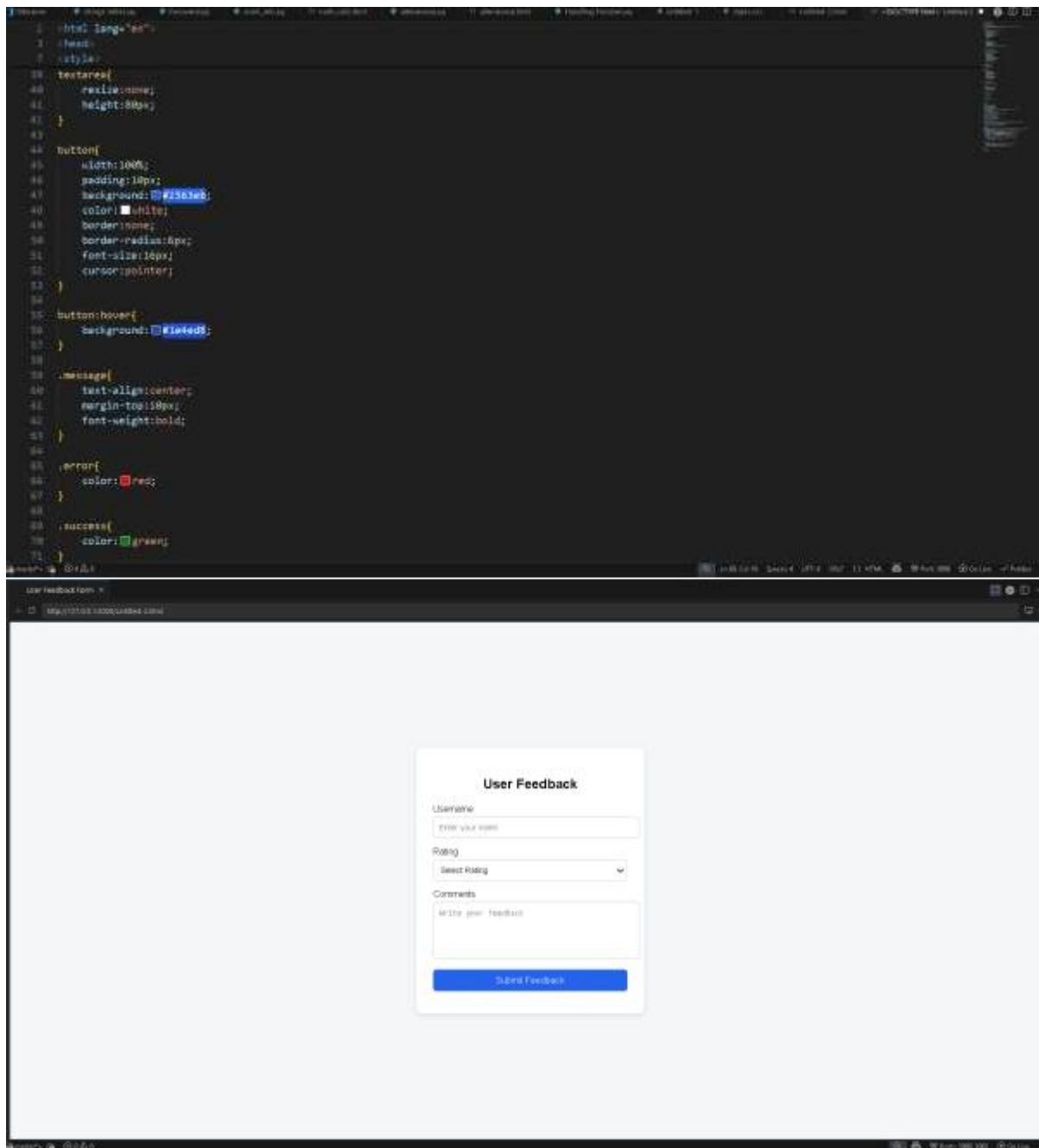
TaskDescription3–(UserFeedbackForm)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use A11y to improve form usability and layout.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 </html>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>User Feedback Form</title>
7 <style>
8 body{
9     font-family: Arial, sans-serif;
10    background-color: #f4f6f8;
11    display: flex;
12    justify-content: center;
13    align-items: center;
14    height: 100vh;
15    margin: 0;
16 }
17
18 .container{
19     background: white;
20     padding: 10px;
21     border-radius: 10px;
22     width: 300px;
23     box-shadow: 0 4px 10px #000000;
24 }
25
26 .rating{
27     text-align: center;
28     margin-bottom: 10px;
29 }
30
31 input, select, textarea{
32     width: 100%;
33     padding: 10px;
34     margin-top: 5px;
35     margin-bottom: 10px;
36     border: 1px solid #ccc;
37     border-radius: 5px;
38     font-size: 14px;
39 }
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TaskDescription-4(ImageGallerywithJavaScript)

- Task:Displayasmallimagegalleryonawebpage.

Requirements:

- UseJavaScripttodynamicallyloadimages.
- Add next/previousnavigation buttons.
- UseAltoptimizelayoutresponsiveness.

```

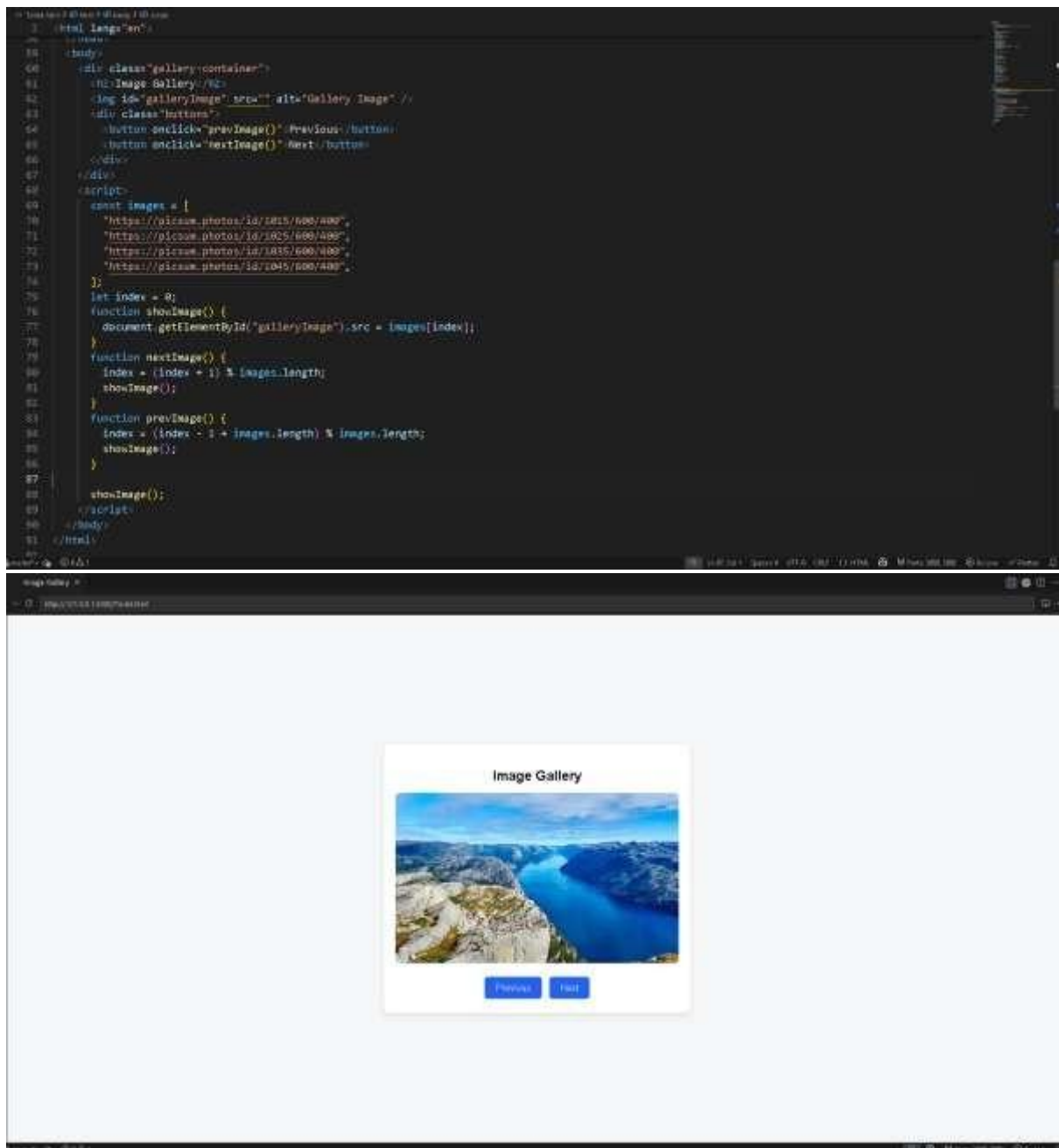
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Image Gallery</title>
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    background: #f4f6f8;
11    margin: 0;
12    display: flex;
13    justify-content: center;
14    align-items: center;
15    height: 100vh;
16  }
17  .gallery-container {
18    text-align: center;
19    background: #ffffff;
20    padding: 10px;
21    border-radius: 10px;
22    box-shadow: 0 4px 10px #000000;
23    max-width: 500px;
24    width: 90%;
25  }
26  h1 {
27    margin-bottom: 15px;
28  }
29  img {
30    width: 100%;
31    height: 300px;
32    object-fit: cover;
33    border-radius: 5px;
34  }
35  .buttons {

```

```

36    margin-top: 15px;
37  }
38  button {
39    padding: 10px 20px;
40    margin: 5px;
41    border: none;
42    border-radius: 5px;
43    background: #f5f5f5;
44    color: #000000;
45    font-size: 16px;
46    cursor: pointer;
47  }
48  button:hover {
49    background: #e0e0e0;
50  }
51  @media (max-width: 500px) {
52    img {
53      height: 200px;
54    }
55  }
56 </style>
57 </head>
58 <body>
59 <div class="gallery-container">
60 <h2>Image Gallery</h2>
61 <img id="galleryImage" src="" alt="Gallery image" />
62 <div class="buttons">
63 <button onclick="prevImage()">Previous</button>
64 <button onclick="nextImage()">Next</button>
65 </div>
66 </div>

```

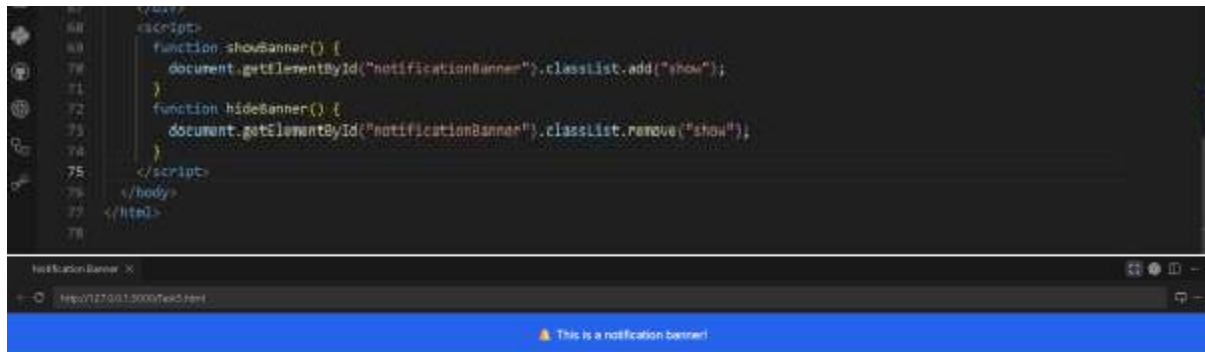
TaskDescription-5:(NotificationBanner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use A11y to ensure smooth transitions and clean UI behavior.

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1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Notification Banner</title>
7 <style>
8 /* Page styling */
9 body {
10 margin: 0;
11 font-family: Arial, sans-serif;
12 }
13 /* Notification banner */
14 .banner {
15 position: fixed;
16 top: -100px;
17 left: 0;
18 width: 100%;
19 background: #156580;
20 color: white;
21 text-align: center;
22 padding: 10px;
23 font-size: 14px;
24 transition: top 0.5s ease;
25 }
26 /* when banner is visible */
27 .banner.show {
28 top: 0;
29 }
30 /* Close button */
31 .close-btn {
32 position: absolute;
33 right: 10px;
34 top: 10px;
35 cursor: pointer;
36 }
37 </style>
38 </head>
39 <body>
40 <div id="notificationBanner" class="banner">
41 <span class="close-btn" onclick="hideBanner()" />
42 </div>
43 <div class="container">
44 <h1>Notification Banner Example</h1>
45 <p>Click the button to display the notification.</p>
46 <button onclick="showBanner()">Show Notification</button>
47 </div>
48 </body>
49 </html>
50 </script>
```



Notification Banner Example

Click the button to display the notification.

Show Notification